THE GLUEING FUNCTOR IS FAITHFUL ON GLUEABLE SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2647-2653 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2655-2683, the complete author proof (27 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_spaces-pushouts.json (pb=1725, stmt 2647-2653, proof 2655-2683).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F9N: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2470–2474. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05K0: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2683–2683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03HK: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2683–2683. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(R \to R', f)$ be a glueing pair, see above. The functor (0F9N) is faithful on the full subcategory of algebraic spaces Y/X glueable for $(R \to R', f)$.*

Proof. Let $f, g : Y \to Z$ be two morphisms of algebraic spaces over X with Y and Z glueable for $(R \to R', f)$ such that f and g are mapped to the same morphism in the category $Spaces(U \leftarrow U' \to X')$. We have to show the equalizer $E \to Y$ of f and g is an isomorphism. Working étale locally on Y we may assume Y is an affine scheme. Then E is a scheme and the morphism $E \to Y$ is a monomorphism and locally quasi-finite, see Morphisms of Spaces, Lemma 03HK. Moreover, the base

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change of $E \to Y$ to U and to X' is an isomorphism. As Y is the disjoint union of the affine open $V = U \times_X Y$ and the affine closed $V(f) \times_X Y$, we conclude E is the disjoint union of their isomorphic inverse images. It follows in particular that E is quasi-compact. By Zariski's main theorem (More on Morphisms, Lemma 05K0) we conclude that E is quasi-affine. Set $B = \Gamma(E, \mathcal{O}_E)$ and $A = \Gamma(Y, \mathcal{O}_Y)$ so that we have an R-algebra homomorphism $A \to B$. Since $E \to Y$ becomes an isomorphism after base change to U and X' we obtain ring maps $B \to A_f$ and $B \to A \otimes_R R'$ agreeing as maps into $A \otimes_R R'_f$. Since A is glueable for $(R \to R', f)$ we get a ring map $B \to A$ which is left inverse to the map $A \to B$. The corresponding morphism $Y = \mathrm{Spec}(A) \to \mathrm{Spec}(B)$ maps into the open subscheme $E \subset \mathrm{Spec}(B)$ pointwise because this is true after base change to U and X'. Hence we get a morphism $Y \to E$ over Y. Since $E \to Y$ is a monomorhism we conclude $Y \to E$ is an isomorphism as desired. $\square$